12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16

17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

P2RY12

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

P2RY12

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CX3CR1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CX3CR1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

TREM2

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TREM2

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

APOE

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

APOE

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

FTL

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

FTL

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

CD83

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD83

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

NFKB1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

NFKB1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD2

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD2

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SKAP1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SKAP1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

CD14

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD14

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD68

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD68

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

GFAP

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

GFAP

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

AQP4

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

AQP4

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

SLC1A2

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SLC1A2

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SLC4A4

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SLC4A4

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

HSP90AA1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

HSP90AA1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

SERPINH1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SERPINH1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

DNAH9

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

DNAH9

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

SPAG17

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SPAG17

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

VIM

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

VIM

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

S100B

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

S100B

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

IL1RAPL1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

IL1RAPL1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

KIRREL3

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

KIRREL3

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

5

MRC1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

MRC1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD163

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD163L1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163L1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

MSR1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

MSR1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

CLDN5

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CLDN5

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

CD38

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD38

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

CD79A

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CD79A

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

IGHM

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHM

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

2

4

6

IGHG1

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHG1

12

5

3

1
20

8

10 2

21

19

4

0
9

11

7

15

23

26

18

31

25 22

17

29

6

14

13

28

32

24 33
30

16

27

− 10

−5

0

5

10

− 15 −10 −5 0 5 10

UMAP_1

UMAP_2

0

1

2

3

4

TSPO

12

5

3

1

20

8

10

2

21

19 4

0

9

11

7

15

23

26
18

31

25

22

17

29

6

14

13

28

32

24

33

30

16

27

− 50

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TSPO